

All My Work

Your saved learning record for Genetic Diversity in Populations.

7 completed activity runs. Required checks and formative practice are recorded separately.

Unfinished work: 0 required check(s), 0 practice run(s). Resume them from their original sections; they are not counted as completed.

Writing and flash-card self-assessments are not automatically graded. Completion is not a mastery certificate.

Same allele frequencies, different genotype pattern

9/17/2026, 8:41:47 PM • Elapsed 0m 0s • Formative practice

Review submitted responses and attempts

Label A

Attempt 1: Heterozygote deficit — Correct

Reference: Heterozygote deficit

Label B

Attempt 1: Same allele frequencies — Correct

Reference: Same allele frequencies

Combine allele pools using their contributions

9/17/2026, 8:41:47 PM • Elapsed 0m 0s • Formative practice

Review submitted responses and attempts

Label A

Attempt 1: Resident contribution — Correct

Reference: Resident contribution

Label B

Attempt 1: Immigrant contribution — Correct

Reference: Immigrant contribution

Label C

Attempt 1: $p = 0.35$ — Correct

Reference: $p = 0.35$

Bottleneck and founder effect: different histories

9/17/2026, 8:41:47 PM · Elapsed 0m 0s · Formative practice

Review submitted responses and attempts

Label A

Attempt 1: Bottleneck — Correct

Reference: Bottleneck

Label B

Attempt 1: Founder effect — Correct

From recessive phenotype to expected counts

9/17/2026, 8:41:46 PM · Elapsed 0m 0s · Formative practice

Review submitted responses and attempts

Label A

Attempt 1: Observed aa fraction — Correct

Reference: Observed aa fraction

Label B

Attempt 1: Recessive allele frequency — Correct

Reference: Recessive allele frequency

Label C

Attempt 1: Other allele frequency — Correct

Reference: Other allele frequency

Label D

Attempt 1: Homozygous dominant frequency — Correct

Reference: Homozygous dominant frequency

Label E

Attempt 1: Heterozygote frequency — Correct

Reference: Heterozygote frequency

Label F

Attempt 1: Expected carrier count — Correct

Reference: Expected carrier count

Random pairing from an allele pool

9/17/2026, 8:41:46 PM · Elapsed 0m 0s · Formative practice

Review submitted responses and attempts

Label A

Attempt 1: p^2 — Correct

Reference: p^2

Label B

Attempt 1: pq — Correct

Reference: pq

Label C

Attempt 1: pq — Correct

Reference: pq

Label D

Attempt 1: q^2 — Correct

Reference: q^2

Count individuals and allele copies separately

9/17/2026, 8:41:46 PM · Elapsed 0m 0s · Formative practice

Review submitted responses and attempts

Label A

Attempt 1: AA genotype — Correct

Reference: AA genotype

Label B

Attempt 1: Aa genotype — Correct

Reference: Aa genotype

Label C

Attempt 1: aa genotype — Correct

Reference: aa genotype

Label D

Attempt 1: Two allele copies per individual — Correct

Reference: Two allele copies per individual

Populations, gene pools and genetic variation

9/17/2026, 8:41:44 PM · Elapsed 0m 0s · Required check

Review submitted responses and attempts

First-try multiple choice: 1/2. Writing is not automatically graded.

Which observation directly indicates microevolution at a locus?

Options: A change in allele frequency across generations · The same allele proportions occur in a larger sample · Plants differ in height after receiving different amounts of water · One individual gains body mass

Attempt 1: The same allele proportions occur in a larger sample — Incorrect

Attempt 2: A change in allele frequency across generations — Correct

Why must the population boundary be defined?

Options: It guarantees Hardy–Weinberg equilibrium · It determines which individuals and allele copies belong in the sample · It makes all individuals genetically identical · It prevents any future mutation

Attempt 1: It determines which individuals and allele copies belong in the sample — Correct

Use an example to distinguish an allele count from an allele frequency.

Real-origin saved Chapter 19 work. If A occurs 20 times among 100 copies, its frequency is 0.20. If it later occurs 40 times among 200 copies, its count increased but frequency remains 0.20. Frequency compares copies with the relevant total.

Submitted writing; human review required.

Explain why a change in an individual's phenotype does not automatically demonstrate population evolution.

Development or environmental conditions can change a phenotype without altering inherited allele frequencies. Population evolution must be evaluated across the defined population and generations, using evidence of changes in allele proportions.

Submitted writing; human review required.

Frayer work

No Frayer words selected.

Collected optional responses

No optional written responses collected.

Textbook question work

Draft and saved question responses. Optional and ungraded.

In Text · p. 680 · question 1 — Saved

Real-origin saved Chapter 19 work. If A occurs 20 times among 100 copies, its frequency is 0.20. If it later occurs 40 times among 200 copies, its count increased but frequency remains 0.20. Frequency compares copies with the relevant total. Edited.

Supported and independent practice